

EF5342 Financial Systems, Markets and Instruments

Week 10 — Commercial Banks and Investment Banks

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Commercial banks

- Commercial bank activities
- Structure and trends of the US banking industry
- Regulation of commercial banks

Investment banks

- Industry structure and competitive dynamics
- Investment bank activities
- Regulation of investment banks

Commercial Banks

A bank **intermediates between the borrower and the lender**: instead of raising capital directly, a company can borrow from a bank. Banks matter especially in markets where bond and equity markets are less developed.

- **Screen and monitor** borrowers — ensures capital is put to good, productive use
- **Maturity intermediation** — borrow short-term to fund long-term assets
- Play a key role in the **transmission of monetary policy**

Banks are heavily regulated to protect against disruptions to the services they perform.

Multiple Roles of Banks

Lender Banks originate and hold loans (mortgages, business loans, consumer credit, syndicated facilities)

Borrower Banks borrow from depositors, other banks (interbank market), central banks (discount window), or wholesale markets (repos, commercial paper, bonds)

Liquidity providers Lines of credit that firms can draw when needed

Credit enhancement Letters of credit guaranteeing that a buyer's payment to a seller will be received on time and for the correct amount

Payment channel The backbone of payment systems — checks, wires (SWIFT, Fedwire), cards

Conduit for monetary policy Banks transmit central bank policy — lending rates, reserve requirements, and open-market effects flow through banks to the economy

Trustees Act as trustees in structured finance — managing collateral, enforcing terms, distributing cash flows (securitizations, indenture trustees)

Custody agents Safeguard assets — holding securities, cash, or instruments for clients (institutional investors, mutual funds, central counterparties)

Historical Development of the US Banking Industry

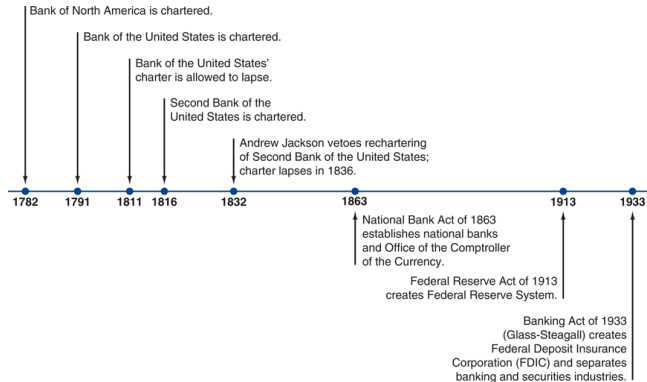


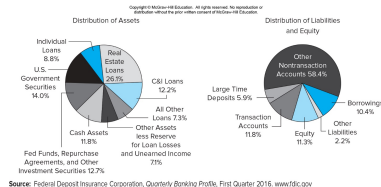
Figure: Timeline of the early history of commercial banking in the United States

Balance Sheets: Commercial Banks vs. Nonfinancial Firms

A bank's balance sheet is typically the **opposite** of a non-financial firm's: a bank's *assets* are loans and its *liabilities* are deposits; a non-financial firm's assets are deposits and its liabilities are loans.

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Commercial Banks		Nonfinancial Firms	
Assets	Liabilities and Equity	Assets	Liabilities and Equity
Loans	Deposits	Deposits	Loans
Other financial assets		Other financial assets	
Other nonfinancial assets	Other liabilities and equity	Other nonfinancial assets	Other liabilities and equity



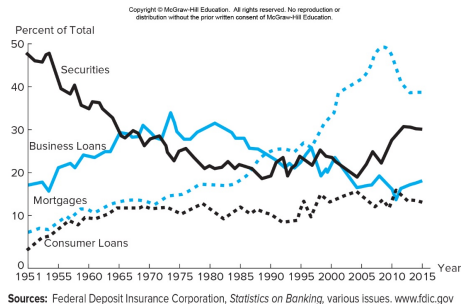
Commercial Bank Assets

Loans generate the most revenue:

- Commercial & industrial (C&I) loans are *declining* — nonbank lending substitutes such as commercial paper
- Mortgages (residential real estate) and consumer loans (credit cards, auto, personal) have increased in importance over the long term

Other assets:

- **Investment securities:** funds lent to other banks, or investments in other debt/equity instruments
- **Cash assets:** held to meet reserve requirements and provide liquidity
- **Other assets:** premises and equipment, other real estate owned, etc.



Commercial Bank Liabilities and Equity

Liabilities Interest-bearing deposits (savings), non-interest-bearing deposits (current accounts), time deposits (fixed deposits), and other borrowings (fed funds, interbank borrowing, repos). Liabilities usually have **shorter maturities and are more liquid** than bank assets — banks face **maturity mismatch and liquidity risk**: if short-term rates rise suddenly, funding costs rise while interest income from assets stays the same

Equity Common and preferred stock, surplus (additional paid-in capital), and retained earnings. Regulators require **minimum equity capital** as a buffer against losses. If deposit funding is cheap, banks keep equity at the minimum — putting funds to maximum use to earn profits, at the cost of exposure to insolvency if market conditions worsen suddenly

Off-Balance Sheet Activities

Off-balance sheet (OBS) activities are commitments or exposures that do not appear as assets or liabilities on the balance sheet under standard accounting rules. They generate fee income, service clients, manage risk, or create contingent obligations without immediately using balance sheet capacity. Major categories:

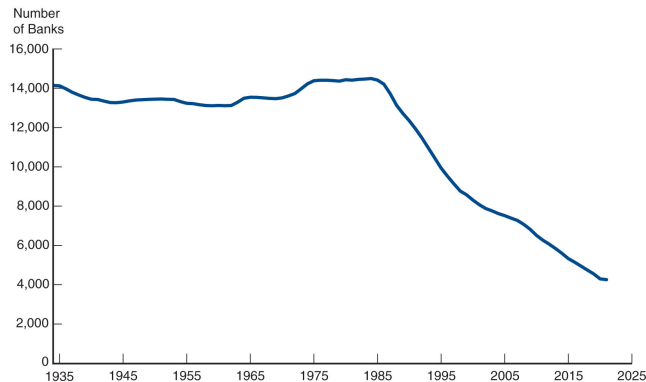
Unused loan commitments Promises to lend in the future (e.g., revolving credit facilities)

Letters of credit Bank charges a fee for guaranteeing a firm's future payment; if the firm defaults, the bank adds the item to liabilities

Loans sold (securitization) The bank may have to buy back loans when quality deteriorates (clawback provisions in the subprime mortgage crisis)

Derivative contracts When speculative bets turn out wrong, the bank could face insolvency

Number of Insured Commercial Banks in the United States, 1934–2022



Source: FDIC, HSOB (chart vintage: year-end 2022).

Current total: 4,238 FDIC-insured institutions as of Q2 2026 (FDIC QBP) — the consolidation trend continues.

Size Distribution of US Banking Institutions

Assets	Number of banks	Share of banks (%)	Share of assets held (%)
Less than \$100 million	543	12.8	0.1
\$100 million–\$1 billion	2,647	62.5	3.9
\$1 billion–\$10 billion	890	21.0	9.4
\$10 billion–\$250 billion	140	3.3	22.9
More than \$250 billion	18	0.4	63.7
Total	4,238	100.0	100.0

FDIC-insured commercial banks and savings institutions, as of Q2 2026 (FDIC QBP; total assets \$26.5 trillion) — Quarterly Banking Profile

Restrictions on Branching

Branching restrictions (McFadden Act of 1927) — very anti-competitive:

- Explicit prohibition on **interstate branching**: national banks could branch only within the state where headquartered

Responses to the restrictions:

Bank holding companies (BHCs) Allowed purchases of banks outside the state; the Fed allowed BHCs a wider scope of activities; BHCs are the dominant corporate form ($\approx 97\%$ of US banking-industry assets are held inside holding-company structures, FDIC Q2 2026)

Automated teller machines (ATMs) Not considered a branch of the bank, so networks were allowed

Bank Consolidation and Nationwide Banking

Why consolidation?

- Loophole mining reduced the effectiveness of branching restrictions; development of super-regional banks
- Economies of scale (increased with information technology); economies of scope in using data for pricing, new products — birth of large, complex banking organizations (LCBOs)
- **Riegle-Neal Act of 1994:** allows full interstate branching — facilitating consolidation, larger institutions, and competition, but raising systemic-risk and local-service concerns
- **Future:** more like other countries, but not quite — several thousand (not several hundred) banks; only half of small banks remain, large banks expected to double in number

Is it a good thing?

Cons:

- Fear of decline of small banks and small-business lending
- Rush to consolidation may increase risk taking

Pros:

- Increases competition and efficiency
- Greater diversification of bank loan portfolios lessens the likelihood of failures

Largest Banks in the United States

Top 10 US banks by total assets (June 30, 2026, FDIC Call Reports):

#	Bank	Total assets
1	JPMorgan Chase Bank, N.A.	\$4.09 trillion
2	Bank of America, N.A.	\$2.65 trillion
3	Citibank, N.A.	\$1.98 trillion
4	Wells Fargo Bank, N.A.	\$1.91 trillion
5	Goldman Sachs Bank USA	\$0.76 trillion
6	U.S. Bank, N.A.	\$0.71 trillion
7	Capital One, N.A.	\$0.66 trillion
8	PNC Bank, N.A.	\$0.61 trillion
9	Truist Bank	\$0.55 trillion
10	BNY Mellon, N.A.	\$0.43 trillion

Bank Performance Measures

Interest rate spread Difference between lending and deposit rates

Net interest margin Interest income minus interest expense, divided by earning assets

Net non-interest margin Non-interest income minus non-interest expense, divided by earning assets (trading profits, service charges on deposits, credit card fees)

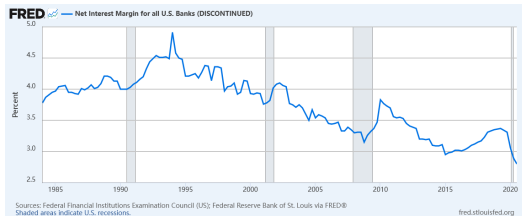
Net charge-offs Percentage of loans written off as uncollectable

Nonperforming loan (NPL) Borrowed money on which the debtor has not made scheduled payments for at least 90 days

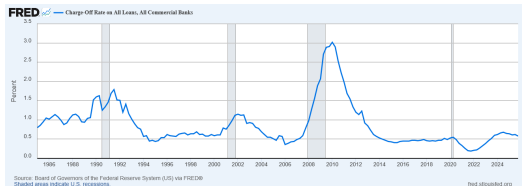
ROA / ROE Net income divided by assets / by equity

Bank profitability declines and charge-offs rise during economic recessions.

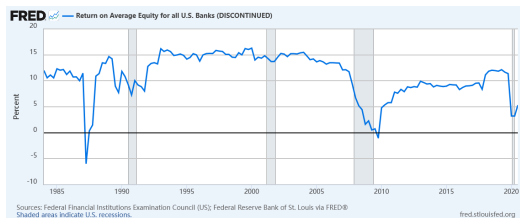
Bank Performance over Time



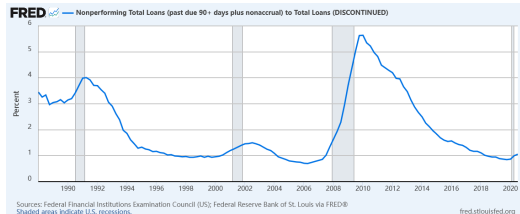
Net interest margin



Charge-off rate



Return on assets



Nonperforming loans

Source: Federal Reserve Bank of St. Louis

What Are the Major Risks Banks Face?

Commercial banks face risks that can threaten their solvency, profitability, reputation, and operational continuity:

- ➊ **Credit risk** — borrowers defaulting on loans or other credit exposures
- ➋ **Market risk & interest rate risk in the banking book** — losses from adverse moves in interest rates, equity prices, FX, or commodities (trading books, asset–liability mismatches)
- ➌ **Liquidity risk** — inability to meet obligations without unacceptable losses (funding or market liquidity)
- ➍ **Operational risk (incl. cyber & IT)** — losses from inadequate or failing processes, people, systems, or external events
- ➎ **Regulatory & compliance risk** — failure to meet laws, rules, or standards, leading to fines, restrictions, or forced changes

US Banking Regulations

FDIC Insures deposits of commercial banks — \$250,000 per bank account; examines and supervises about 4,000 banks for safety and soundness. (*Video: importance of the FDIC for financial stability*)

Dual banking system Banks can be nationally or state-chartered: the Office of the Comptroller of the Currency (OCC) charters and regulates national banks; state agencies charter and regulate state banks

Federal Reserve System Regulatory power over nationally chartered banks and their holding companies, and state banks that opt into the Federal Reserve System

International Banking Regulations: Basel III

Basel III — a comprehensive international framework from the Basel Committee on Banking Supervision (BCBS) to strengthen global banking resilience: more and better capital, less leverage, better liquidity management, enhanced risk coverage.

Pillar 1: Minimum capital requirements Higher quantity and quality of capital, emphasizing **Common Equity Tier 1 (CET1)**; minimum CET1 ratio raised to **4.5% of risk-weighted assets**

Pillar 2: Supervisory review Banks assess their overall risk profile; supervisors review it (SREP) — promoting stronger governance, stress testing, and risk management

Pillar 3: Market discipline Detailed, standardized public disclosures on capital, risk exposures, RWAs, leverage, liquidity, and remuneration — so markets can assess bank soundness

The most significant overhaul of banking regulation since the 1930s.

Investment Banks (IBs)

Investment banking: financial intermediation services helping corporations, governments, and institutional investors **raise capital, execute strategic transactions, and manage financial risk**. Principal business lines:

M&A advisory Buy-side and sell-side counsel on corporate transactions, valuations, and deal structures

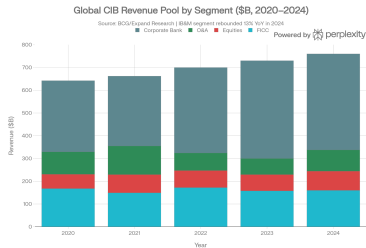
Equity capital markets (ECM) Underwriting and distributing IPOs, follow-on offerings, convertibles

Debt capital markets (DCM) Underwriting investment-grade and high-yield bonds, leveraged loans, syndicated credit facilities

Sales & trading Intermediating client transactions and managing positions in equities and FICC

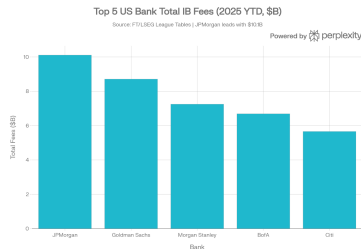
Research Equity and credit analysis for institutional investors

Market Size and Industry Structure



Global IB revenue trends

Investment banking is a **highly concentrated oligopoly**: per FT/LSEG league tables, YTD 2026 (H1 global IB fees \$78 billion, +14% year-over-year), the top five US banks collectively dominate the global IB fee pool.



Investment Banking Scorecard

Global IPOs Bookrunner Ranking					
YTD 2025			YTD 2024		
Bookrunner	Value \$bn	#	Rank	Value \$bn	#
Morgan Stanley	7.37	66	1	5.99	56
Cantor Fitzgerald & Co	7.18	34	7	3.30	19
JPMorgan	5.65	51	2	5.81	52
Goldman Sachs	5.13	55	3	5.07	39
Citi	5.05	47	5	4.27	42
BTIG LLC	4.23	25	20	1.35	7
BoTA Securities	4.09	40	4	4.83	42
Cohen & Co Inc	4.03	28	47	0.50	6
CITIC Securities	3.85	43	15	1.53	31
Jefferies LLC	3.09	38	9	2.68	32
Subtotal	49.79	236		35.32	178
Total	155.11	1,237		111.89	1,153

Global M&A Advisor Ranking					
YTD 2025			YTD 2024		
Advisor	Value \$bn	#	Rank	Value \$bn	#
Goldman Sachs	1,491.9	465	1	866.4	385
JPMorgan	1,103.6	399	2	684.3	364
Morgan Stanley	1,035.2	339	3	638.6	331
BoTA Securities	712.3	271	5	398.3	285
Citi	570.7	217	4	510.9	204
Evercore Inc	415.9	201	6	294.7	176
Centerview Partners	380.7	102	9	256.3	97
Barclays	348.0	145	8	260.6	161
Wells Fargo	342.3	61	16	157.6	66
UBS	282.9	198	7	273.6	214

Source: Dealogic Investment Banking Scorecard

Bulge bracket banks — the designation derives from listing the largest underwriters first (in larger, “bulging” type) on tombstone announcements. As of 2026, eight firms qualify: JPMorgan, Goldman Sachs, Bank of America, Morgan Stanley, Citigroup, Barclays, UBS, and Deutsche Bank. Defining characteristics:

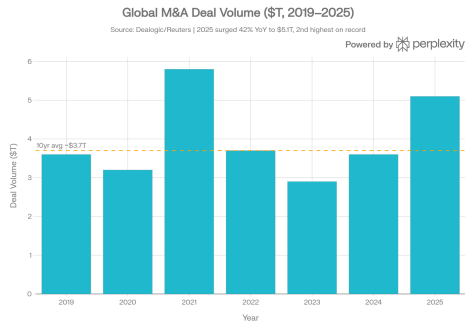
- Global presence across 60+ countries and 100+ markets
- Full-service capability: M&A, ECM, DCM, trading, research, asset management
- Massive balance sheets enabling multi-billion-dollar underwritings and bridge commitments
- Clients: Fortune 500 corporations, sovereign wealth funds, institutional investors

Industry Structure: Firm Typology (2/2)

- Elite boutiques** M&A advisory without the bulge-bracket balance sheet — Evercore, Lazard, Centerview Partners, Moelis, PJT Partners challenge bulge brackets for advisory mandates on landmark deals. Flat hierarchies, senior banker attention, and a conflict-free advisory model appeal to certain clients
- Middle-market banks** Jefferies, Houlihan Lokey, William Blair, Piper Sandler — serving companies in the \$50M–\$500M revenue range
- Regional and specialty boutiques** Focus on specific geographies (Southwest US, Pacific Northwest) or sectors (healthcare, energy, technology); particularly active in transactions below \$100M and critical training grounds for early-career bankers

Mergers & Acquisitions Advisory

- **Fees:** typically 1–2% of deal value for smaller transactions, tapering to 0.1–0.5% for mega-deals (>\$5B)
- **Buy-side vs. sell-side:** advising acquirers on identifying and valuing targets; advising targets on auction processes and defence
- **Fairness opinions:** independent valuations for boards facing conflicts of interest
- **Fairness-of-price analysis:** DCF models, LBO analysis, trading comparables, precedent transactions — the core financial modeling toolkit



Deal Valuation Methodologies

Investment bankers regularly employ three core approaches — always used in combination (a “football field”):

Discounted cash flow (DCF) Projects free cash flows and discounts them at the WACC

Comparable company analysis (“comps”) Applies trading multiples (EV/EBITDA, P/E, P/S) from public peers to the target; adjustments for size, growth, and liquidity are critical

Precedent transaction analysis Uses acquisition premiums and multiples from prior M&A deals; typically yields *higher* valuations than comps due to control premiums (historically averaging 25–35%)

Equity Capital Markets (ECM)

ECM groups bridge corporate clients seeking equity financing with institutional investors.

- ECM bankers manage the **book-building** process — building an order book from institutional investors — and price shares in a **roadshow**
- Relationships with buy-side accounts (mutual funds, hedge funds, pension funds) and the ability to manage price discovery are critical competitive advantages
- Two methods for the IPO process: **best efforts** and **firm commitment** (worked example next)

Best Efforts vs. Firm Commitment: Example (1/3)

An investment bank underwrites **20,000,000 shares** of Murray Construction Corp. on a **firm commitment** basis: the bank pays \$15.50 per share to Murray and sells to the public at \$16.35. **Profit to the bank?**

$$\text{Profit} = (\$16.35 - \$15.50) \times 20,000,000 = \$17,000,000$$

From Murray Construction's perspective, the \$17,000,000 is the **commission** it pays to issue the stock.

Best Efforts vs. Firm Commitment: Example (2/3)

Firm commitment downside: if the bank can only sell at \$14.75:

$$\text{Murray receives } \$15.50 \times 20,000,000 = \$310,000,000$$

$$\text{Bank profit} = (\$14.75 - \$15.50) \times 20,000,000 = -\$15,000,000$$

— the **risk** of the firm commitment.

Best efforts instead: the bank sells 18,400,000 shares at \$15.50 and charges Murray \$0.375 per share sold. At \$15.50:

$$\text{Murray receives } (\$15.50 - \$0.375) \times 18,400,000 = \$278,300,000$$

$$\text{Bank profit} = \$0.375 \times 18,400,000 = \$6,900,000$$

If shares sell at only \$14.75: Murray receives $(\$14.75 - \$0.375) \times 18,400,000 = \$264,500,000$ — but the bank's profit is *still* \$6,900,000.

Debt Capital Markets (DCM)

DCM covers multiple markets simultaneously:

Investment-grade corporate bonds Lower yield, tighter spreads; dominated by BofA, JPMorgan, Citi

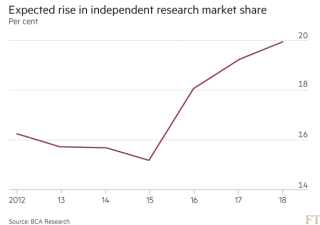
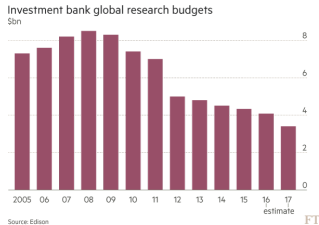
High-yield (“junk”) bonds Higher yield; issuance rebounded strongly in 2024–2025 after the 2022–23 drought

Leveraged loans / CLOs Floating-rate instruments integral to leveraged buyouts

ABS and CMBS Securitized pools of mortgages, auto loans, or commercial real estate

Banks sit at the intersection of trading floors and corporate coverage — understanding investor demand and issuer needs simultaneously.

- Help execute trades of securities; provide investment research to clients (e.g., analyst reports)
- **Prime brokerage:** helping hedge funds trade, get leverage, locate shares for shorting, advise on trading ideas



Research — and Trading Revenues

Research sits at the center of an information network: serving *clients* (institutional investors), the *media*, *information providers*, and drawing on *industry sources*, alongside CFE (certified financial examiners) and client interactions.

“Flow” trading Client-driven revenues — fees, commissions, markups, and spreads measured through sales and production credits

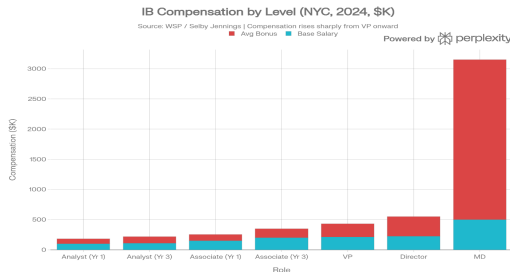
“Prop” trading Risk-driven revenues — dedicated proprietary trading delinked from customer business

For most players, capital markets are primarily a *client* business, not a proprietary trading business — client franchise building is critical. The **Volcker rule** (part of the Dodd-Frank Act) prohibits prop trading in banks in the US.

Career Paths and Compensation

Investment banking offers some of the highest entry-level compensation in any profession; the hierarchy is steep, standardized, and performance-driven.

- Typical path: **Analyst** → **Associate** → **VP** → **Director/ED** → **Managing Director**
- Most analysts join from undergraduate programs, spend 2–3 years, then exit to private equity, hedge funds, or an MBA with re-entry at Associate level



Technology and AI Transformation

AI is reshaping investment banking from front-office deal origination to back-office compliance:

- Real-time transaction monitoring and fraud detection (JPMorgan reports a 20% reduction in payment validation rejection rates)
- Automated credit underwriting and due diligence
- Equity research synthesis and earnings analysis
- Client segmentation and pitch book generation
- Risk model construction and regulatory reporting

AI does not eliminate the investment banker; it eliminates the **junior tasks** that once justified entry-level positions. Develop judgment on complex, ambiguous problems, client relationship skills, and the ability to critically interpret AI outputs — not just Excel modeling.

Current Regulatory Environment

- Post-2008, the Basel III framework dramatically increased capital requirements — banks now maintain CET1 ratios of no less than 12%
- **Basel III Endgame re-proposed (March 19, 2026)**: the July 2023 proposal is dead; the re-proposal applies the expanded risk-based approach only to the largest banks (Category I–II, ≈ 8 US G-SIBs) and would *reduce* aggregate CET1 requirements for those firms by $\approx 4.8\%$ (comment period closed June 18, 2026; final rule targeted for end-2026)
- The **eSLR** was finalized separately and took effect April 1, 2026 — recalibrated to 50% of the Method 1 G-SIB surcharge (bank-subsidary cap 1%), restoring it as a backstop rather than a binding constraint
- **G-SIB surcharge**: reform proposed March 2026 (not final); 29 G-SIBs worldwide in the FSB 2025 list, 8 of them US — surcharge buckets 1.0%–2.5% (JPMorgan highest at 2.5%)

Investment Banks & Commercial Banks

- Investment banks were essentially created in the US by the **Glass-Steagall Act (1933)**; before that, investment banking activities were part of large, money-center commercial banks
- As a result, J.P. Morgan & Co. was broken up — its investment banking activities spun off into **Morgan Stanley**
- The lines between investment and commercial banks began to blur again in the 1980s
- After the **Citi-Travellers merger (1998)**, the **Gramm-Leach-Bliley Act (1999)** (Financial Services Modernization Act) effectively repealed Glass-Steagall

Framework for the Banking & Securities Industry

Universal banking Banks provide a full range of financial services (insurance, securities, real estate) under the same legal entity

Legal separation Banks are prohibited from engaging in other financial services such as securities issuance and insurance

Financial supermarket (FHCs/BHCs) Full range of services, but each activity under a different legal entity (subsidiary)

Global banks Universal banking is more prevalent in Europe than the US — e.g., Bank of America: commercial banking (BoFA, N.A.), investment banking (BAML), wealth management (Merrill Lynch Wealth Management)

Should Glass-Steagall Be Reinstated?

Pros

- Protect depositors
- Simplify regulation of depository institutions
- More transparency for depository institutions' customers
- Lessens systemic risk and limits financial-risk contagion

Cons

- Restricts financial innovation
- Limits profit opportunity and competition
- Banks will find other loopholes, just like in the past

Key Takeaways

- **Commercial banks:** intermediaries taking deposits (liabilities) and issuing loans (assets); profit from the interest rate spread but face maturity mismatch and liquidity risk
- **Bank regulation:** heavy regulation for stability and depositor protection — FDIC (deposit insurance), the Fed, the OCC; minimum equity capital buffers losses
- **Investment banks:** operate primarily in capital markets — underwriting new securities (debt/equity) and M&A advisory; revenue from fees, commissions, and trading spreads
- **Two ways to underwrite:** firm commitment — the bank bears the risk of unsold shares (buys the entire issue); best efforts — the issuer bears the risk, the bank earns a commission on shares sold
- **Industry evolution:** Glass-Steagall (1933) separated commercial and investment banking; effectively repealed in 1999, enabling financial conglomerates (BHCs, FHCs) offering both under one umbrella